

Harish Tirumalai

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EDUCATION

Georgia Institute of Technology

Expected August 2026

Bachelor of Science in Computer Science, Minor in Physics

- Concentrations: Intelligence and Theory
- Major Coursework: Data Structures and Algorithms, Advanced Algorithms, Intro to AI, Machine Learning, Deep Learning, Discrete Mathematics, Probability and Statistics, Linear Algebra I & II, Object Oriented Programming
- Minor Coursework: Physics I-III, Quantum Mechanics I, Quantum Information & Quantum Computing

EXPERIENCES

Georgia Tech Research Institute

May 2025 - Present

Machine Learning Intern

Atlanta, GA

- Enabled 100% visual validation of a detection algorithm by parsing raw protobuf streams & render the resulting detections
- Automated $\geq 80\%$ of manual annotation by implementing a local search algorithm for label propagation across frames
- Integrated Point Cloud Models into an LLM Guided Evolution framework cited by **Google DeepMind's AlphaEvolve**

Automated Algorithm Design

January 2024 - May 2025

Genetic Programming Research

Atlanta, GA

- Boosted model accuracy by 7% by integrating **PointNet++** into an LLM-driven architectural augmentation pipeline
- Deployed a local model-serving API to enable rapid, iterative augmentation of PointNet++ architectures by an LLM
- Architected an LLM-based architecture generator (LLMGE) with robust error handling to prevent invalid model creation

PROJECTS

Enhanced Image Annotation Tool

June 2025 - July 2025

- Overhauled the streamlit-image-label library with core features like bounding box creation, multi-deletion, and labeling
- Streamlined the annotation workflow with key UX features like a zoom-on-select and auto-resizing for boxes $\leq 64 \times 64$ pixels
- Modernized the data pipeline by switching from XML to CSV I/O, boosting compatibility with data science libraries

HumTune AI

January 2024 - December 2024

- Consistently achieved $\geq 94\%$ accuracy in hum-to-song matching using a **CNN** trained on sine wave audio representations
- Developed a responsive web application interface using React for recording hummed melodies and viewing song matches
- Reduced song identification to < 5 seconds by optimizing the CNN architecture and audio feature extraction pipeline

Dungeon Crawler

August 2023 - December 2023

- Implemented Strategy and Observer patterns to build a modular collision system with map-specific validation rules
- Designed a scalable enemy generation system using the Factory Method pattern to spawn map-specific enemy types
- Ensured a globally accessible game state for scoring by implementing the Singleton pattern for the leaderboard system

LEADERSHIP EXPERIENCE

Research Team Lead

- Led a team of 10 researchers on a project focused on LLMGE, delegating tasks, setting project milestones, managing deadlines to ensure the timely completion of objectives, and reporting improved performance of previously obtained metrics
- Onboarded and mentored junior team members by providing comprehensive guidance on project background, environment setup, and strategic objectives to accelerate their integration and contribution to the team

Georgia Tech Solar Racing

- Directed a team of 2 software engineers in developing a real-time data platform to visualize and simulate vehicle telemetry
- Developed a curve detection algorithm using sliding window, achieving 91% accuracy to optimize vehicle race performance
- Improved battery consumption model for race simulations, by incorporating slope-adjusted speeds to improve energy usage
- Implemented an interactive dashboard to provide optimal speed recommendations by visualizing live GPS curve analysis

TECHNICAL SKILLS

Front End: Angular, HTML, JavaScript, React

Back End: Java, Python, C/C++, SQL, Linux

Developer Tools: Github, Gitlab, Docker, VS Code, Xcode, PyCharm, IntelliJ

Libraries: Scipy, Pandas, NumPy, SciKit, PyTorch, TensorFlow, Hugging Face Transformers, Matplotlib, DEAP, Streamlit